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Module 6 — Input/Output, Variables, Arrays, Function, and Subroutine

Learning outcomes

Students will be able to:

- read and write data with Cells, Range, InputBox, and MsgBox;
- declare variables with appropriate types and scope;
- use an array for a collection of values;
- distinguish a Function from a Sub; and
- build a modular program that processes a table.

100-minute sequence

Minutes	Activity
0–10	Identify IPO in the previous linear macro
10–25	Cells, Range, InputBox, and MsgBox
25–40	Variable types and local/public scope
40–55	One- and two-dimensional arrays
55–70	Difference between Function and Sub
70–85	Process a volume table
85–97	Three-question short exercise
97–100	Compile and complete checklist

Minimum product: one volume function and one Sub that reads at least five rows into an array, processes them, and writes results to Excel.

1. Simple input and output

Range and Cells

Dim value **As** Double

```
value = Range("B2").Value  
Cells(2, 3).Value = value * 2
```

Range("B2") is readable for a fixed address. Cells(row, column) is useful when the row or column changes inside a loop.

Prefer an explicit worksheet:

```
Dim ws As Worksheet
Set ws = ThisWorkbook.Worksheets("VolumeData")
```

```
value = ws.Range("B2").Value
ws.Cells(2, "F").Value = value
```

InputBox and MsgBox

```
Sub InputOutputDemo()
    Dim projectName As String

    projectName = InputBox("Enter the project name:")

    If projectName = "" Then
        MsgBox "No project name was entered.", vbExclamation
    Else
        MsgBox "Project: " & projectName, vbInformation
    End If
End Sub
```

A standard InputBox returns text. Use IsNumeric before converting numeric text with CDb1.

2. Variables and data types

```
Dim length_m As Double
Dim segmentCount As Long
Dim segmentName As String
Dim dataIsValid As Boolean
Dim surveyDate As Date
```

Use:

- Double for fractional engineering quantities;
- Long for counts and row indices;
- String for names and codes;
- Boolean for true/false state; and
- Date for dates.

Option Explicit requires declarations and exposes misspelled variable names.

3. Local and public scope

A local variable is declared inside a procedure and is available only there:

```
Sub LocalExample()
    Dim total_m3 As Double
    total_m3 = 10
End Sub
```

A public variable is declared at the top of a standard module:

Option Explicit

```
Public activeProjectName As String
```

Prefer local variables. A public variable is justified only when state must genuinely be shared among modules or forms; its value is harder to trace because several locations may change it.

4. Arrays

An array stores several values of the same kind under one name.

```
Sub ArrayDemo()  
    Dim volumes_m3(1 To 3) As Double  
    Dim i As Long  
    Dim total_m3 As Double  
  
    volumes_m3(1) = 2.5  
    volumes_m3(2) = 3.5  
    volumes_m3(3) = 4  
  
    For i = LBound(volumes_m3) To UBound(volumes_m3)  
        total_m3 = total_m3 + volumes_m3(i)  
    Next i  
  
    MsgBox "Total = " & total_m3 & " m³"  
End Sub
```

An Excel block can be loaded directly into a two-dimensional Variant array:

```
Dim data As Variant  
data = ws.Range("B2:D6").Value  
  
' data(1,1) is B2; data(1,2) is C2; data(1,3) is D2
```

5. Function and Subroutine

A Function accepts parameters and returns a value:

```
Public Function RectangularVolume( _  
    ByVal length_m As Double, _  
    ByVal width_m As Double, _  
    ByVal height_m As Double) As Double  
  
    If length_m <= 0 Or width_m <= 0 Or height_m <= 0 Then  
        RectangularVolume = -1  
    Else  
        RectangularVolume = length_m * width_m * height_m  
    End If  
End Function
```

A Sub coordinates actions: reading a table, calling the function, and writing output.

```
Public Sub CalculateVolumeTable()  
    Dim ws As Worksheet  
    Dim data As Variant  
    Dim results() As Variant  
    Dim i As Long
```

```

Dim volume_m3 As Double

Set ws = ThisWorkbook.Worksheets("VolumeData")
data = ws.Range("B2:D6").Value
ReDim results(1 To UBound(data, 1), 1 To 2)

For i = 1 To UBound(data, 1)
    If IsNumeric(data(i, 1)) And IsNumeric(data(i, 2)) And _
        IsNumeric(data(i, 3)) Then

        volume_m3 = RectangularVolume(CDbl(data(i, 1)), _
                                         CDbl(data(i, 2)), _
                                         CDbl(data(i, 3)))

        If volume_m3 >= 0 Then
            results(i, 1) = volume_m3
            results(i, 2) = "OK"
        Else
            results(i, 1) = Empty
            results(i, 2) = "Invalid dimensions"
        End If
    Else
        results(i, 1) = Empty
        results(i, 2) = "Not numeric"
    End If
Next i

ws.Range("E2:F6").Value = results
ws.Range("E2:E6").NumberFormat = "0.000"
End Sub

```

VolumeData worksheet

Column	Content
A	Segment
B	Length (m)
C	Width (m)
D	Height (m)
E	Volume (m ³)
F	Status

Include five rows, with one zero dimension and one text value, so validation can be observed.

6. Why modular design matters

RectangularVolume knows nothing about cell locations. It is easy to test and can be reused by the table macro and by the UserForm in Module 7. CalculateVolumeTable coordinates Excel input/output but does not own the geometry formula.

Test the function in the Immediate Window:

```

? RectangularVolume(10, 2, 0.3)
? RectangularVolume(0, 2, 0.3)

```

Expected outputs are 6 and -1.

Short exercise — 3 questions

1. Predict

Variable `total_m3` is declared inside Sub A. Can Sub B read it directly? What redesign is safer than making the variable public?

2. Fix

The third row contains the text two. Identify the condition that prevents `CDbl("two")` from running, and explain what the program writes as output.

3. Explain

Why should the volume formula belong to `RectangularVolume`, while worksheet reading/writing belongs to `CalculateVolumeTable`? State one testing benefit.

1. No. A local variable exists only in its procedure. Pass the value as a parameter or return it from a Function.
2. `IsNumeric` is checked before `CDbl`; the volume is blank and status becomes `Not numeric`.
3. Separating responsibilities keeps the calculation independent of the worksheet. The function can be tested directly with known inputs and expected outputs.

Checklist

- ☐ Every variable is declared with an appropriate type.
- ☐ Input is checked before conversion.
- ☐ I distinguish local and public scope.
- ☐ I distinguish the roles of Function and Sub.
- ☐ All five rows are processed even when one row is invalid.

Further reading

1. Michael Alexander & Dick Kusleika, *Excel 2019 Power Programming with VBA*, Wiley, 2019.
2. Bill Jelen & Tracy Syrstad, *Microsoft Excel VBA and Macros (Office 2021 and Microsoft 365)*, Microsoft Press, 2022.
3. Steve McConnell, *Code Complete*, 2nd ed., Microsoft Press, 2004.
4. Brian W. Kernighan & Rob Pike, *The Practice of Programming*, Addison-Wesley, 1999.
5. E. Joseph Billo, *Excel for Scientists and Engineers: Numerical Methods*, Wiley, 2007.

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